

GRADUATION PROJECT — AI ENGINEERING BOOTCAMP

The Student's Channel Brain

Final Project Summary Report

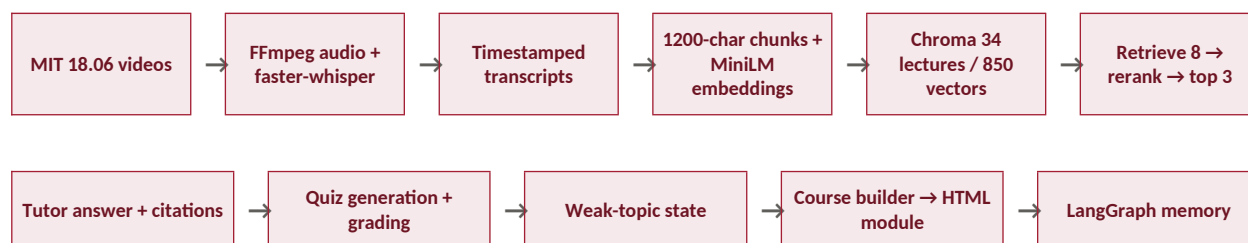
MIT 18.06 Linear Algebra Tutor — end-to-end RAG + agent system

34 Lectures — course coverage	32/38 Final evaluation	84.2% Pass rate
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1. Problem and Business Case

Students may have dozens of hours of lecture recordings but no practical way to ask the recordings a question, jump to the exact supporting moment, or turn weak areas into targeted study material. This project builds a grounded AI tutor over MIT OpenCourseWare 18.06 Linear Algebra. It answers from the real lecture corpus, returns lecture/timestamp citations, generates and grades quizzes, tracks weak topics, and can build a browsable HTML study module from the student's needs.

2. Final Architecture and Data Flow



LangSmith tracing + Gradio UI wrap the full pipeline above.

Production retrieval path: vector search (k=8) → CrossEncoder reranker (ms-marco-MiniLM-L6-v2) → best 3 evidence chunks → tutor/agent.

3. Notebook Responsibilities

Notebook	Role	Primary output
01 — Ingestion	Video → audio → faster-whisper STT; official lecture metadata; validation	Timestamped transcript + metadata files
02 — RAG	Timestamp-aware chunking, all-MiniLM-L6-v2 embeddings, Chroma, reranking	Persistent 34-lecture RAG index
03 — Agent	Tutor tools, quiz generation/grading, weak-topic state, conversational memory	Agent MVP using shared RAG
04 — Web App	Gradio interface, LangSmith tracing, course builder, session helpers	End-to-end product
05 — Benchmark	Controlled 5-lecture configuration benchmark + trace analysis	Benchmark decisions and evidence
06 — Evaluation	Frozen 38-task end-to-end validation and failure analysis	Final system metrics

4. Benchmark Results and Production Configuration

Notebook 05 benchmarked three retrieval decisions on eight fixed questions over Lectures 1–5. The metric was top-lecture correctness, with retrieval latency recorded for each run.

Comparison	Result A	Result B	Decision / interpretation
Chunk size	1200: 62.5%	600: 50.0%	1200 selected; clearest measured quality gain.
Retrieval k	k=3: 62.5%	k=5: 62.5%	No quality gain in the small benchmark.
Reranker	OFF: 62.5% 0.0163 s	ON: 62.5% 0.0444 s	Changed ordering and added ~29 ms reranker time; aggregate accuracy tied.

Final production configuration. Chunk size 1200; initial retrieval k=8; rerank_top_n=3; reranker ON; Lectures 1–34; LLM gpt-5.6-luna with reasoning_effort="high".

5. Final 38-Task Evaluation

38/38 Completed	32 Passed	84.2% Overall pass rate
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Category	Pass	Total	Pass rate
Direct Concept	6	6	100.0%
Lecture-Specific	5	6	83.3%
Comparison	4	5	80.0%
Memory / Follow-up	3	5	60.0%
Out-of-Scope	4	4	100.0%
Quiz Gen/Grading	3	4	75.0%
Weak-Topic Tracking	3	3	100.0%
Course Builder	2	3	66.7%
Source/Timestamp Check	2	2	100.0%

Timestamp/citation correctness: 26/28 applicable checks = 92.9%. Ten tasks were marked N/A because timestamp correctness was not applicable or not recorded. The two dedicated Source/Timestamp Check tasks both passed (2/2).

6. Failure Analysis

Task / area	Observed failure	Engineering interpretation
12 Lecture 33	Pseudoinverse omitted although required by the pass criteria.	Content-completeness / scope failure.
15 Comparison	Used Lecture 22/23 evidence instead of the required foundational Lecture 21 evidence.	Grounding was relevant, but lecture selection was wrong.
21 Memory/follow-up	Transient syntax error; resend succeeded and content was correct.	Runtime reliability issue, not demonstrated memory misunderstanding.
22 Memory/follow-up	SyntaxError interrupted the multi-turn quiz/recall workflow.	Session/runtime robustness issue.
30 Quiz grading	Ambiguous answer was silently graded instead of requesting A-D.	Input-validation / grounding behavior needs stricter guardrails.
36 Course builder	Auto-selection fell back to Lecture 31 instead of the first recorded weak topic, Lecture 21.	Application-state selection failure.

7. Limitations

- The configuration benchmark used only 8 retrieval questions and 5 lectures; it supports local design decisions but is too small for broad system-level claims.
- The benchmark reranker test used $k=3$ and $\text{top}_n=3$, so it measured ordering impact rather than the recall benefit of reranking a larger candidate pool.
- The final system uses audio-only ingestion through FFmpeg + faster-whisper STT; it does not analyze the visual content of lecture videos (slides, whiteboard writing, diagrams).
- The 38-task evaluation is manually judged and remains a finite sample; 10 tasks did not have an applicable/recorded timestamp correctness label.
- Two failures were transient runtime/session errors, showing that reliability under repeated multi-turn use still needs hardening.

8. Future Work

- Harden multi-turn session and tool execution with stricter error handling, retry-safe state updates, and regression tests for quiz/memory flows.
- Add query decomposition for multi-concept comparisons and stronger evidence coverage checks before generating an answer.
- Strengthen course-builder state selection so weak-topic ordering and fallback behavior are deterministic and validated.
- Explore multimodal understanding — sampling slide/whiteboard frames and aligning them with the transcript — to see whether visual evidence improves answers and citations.
- Expand evaluation beyond 38 tasks, automate timestamp validation where feasible, and compare the production $k=8 \rightarrow \text{rerank} \rightarrow \text{top 3}$ design against a simpler baseline on the full 34-lecture corpus.

Final takeaway: The finished system demonstrates end-to-end AI engineering across ingestion, RAG, reranking, agent tools and memory, observability, web deployment, benchmarking, and honest evaluation. The final measured result is 32/38 tasks passed (84.2%), with 92.9% timestamp/citation correctness across applicable checks.